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Oefeningenreeks 7A nr. 10.17

$$\begin{aligned}f(r, s) &= \frac{r^2 - s^2}{r^2 + s^2} \Rightarrow Df(r, s) = \left( \frac{\partial f}{\partial r} \quad \frac{\partial f}{\partial s} \right) \\ \frac{\partial f}{\partial r} &= \frac{2r(r^2 + s^2) - 2r(r^2 - s^2)}{(r^2 + s^2)^2} = \frac{4rs^2}{(r^2 + s^2)^2} \\ \frac{\partial f}{\partial s} &= \frac{-2s(r^2 + s^2) - 2s(r^2 - s^2)}{(r^2 + s^2)^2} = \frac{-4sr^2}{(r^2 + s^2)^2} \\ \Rightarrow Df(r, s) &= \left( \frac{4rs^2}{(r^2 + s^2)^2} \quad \frac{-4sr^2}{(r^2 + s^2)^2} \right)\end{aligned}$$

## References